

HW#1

Homework Report

Course	ECG 782
Student Name	Chang Wang
Student ID	2002721058
Instructor	Ebrahim Saberinia
Date	09-08-2026

Problem 01

The data $\{x[0], x[1], \dots, x[N-1]\}$ are observed where the $x[n]$'s are independent and identically distributed (IID) as $\mathcal{N}(0, \sigma^2)$. We wish to estimate the variance σ^2 as

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n].$$

Is this an unbiased estimator? Find the variance of $\hat{\sigma}^2$ and examine what happens as $N \rightarrow \infty$.

Solution:

Consider the estimator

$$\widehat{\sigma^2} = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n], \quad x[n] \sim \mathcal{N}(0, \sigma^2).$$

To determine whether it is unbiased, note that

$$E[x^2[n]] = \text{Var}(x[n]) + (E[x[n]])^2 = \sigma^2.$$

Therefore,

$$E[\widehat{\sigma^2}] = \frac{1}{N} \sum_{n=0}^{N-1} E[x^2[n]] = \sigma^2.$$

Thus, $\widehat{\sigma^2}$ is an unbiased estimator of σ^2 .

To calculate its variance, use the fourth-moment identity for a zero-mean Gaussian random variable:

$$E[x^4[n]] = 3\sigma^4.$$

It follows that

$$\text{Var}(x^2[n]) = E[x^4[n]] - (E[x^2[n]])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4.$$

Since the observations are independent, their squares are also independent. Hence,

$$\begin{aligned} \text{Var}(\widehat{\sigma^2}) &= \frac{1}{N^2} \sum_{n=0}^{N-1} \text{Var}(x^2[n]) \\ &= \frac{1}{N^2} (N \cdot 2\sigma^4) \\ &= \frac{2\sigma^4}{N}. \end{aligned}$$

As $N \rightarrow \infty$, the variance approaches zero. Since the estimator is unbiased, its mean squared error satisfies

$$E[(\widehat{\sigma^2} - \sigma^2)^2] = \frac{2\sigma^4}{N} \rightarrow 0.$$

Therefore, $\widehat{\sigma^2}$ converges to σ^2 in mean square. Furthermore, by Chebyshev's inequality, for every $\varepsilon > 0$,

$$P(|\widehat{\sigma^2} - \sigma^2| \geq \varepsilon) \leq \frac{2\sigma^4}{N\varepsilon^2} \rightarrow 0.$$

Thus, the estimator is consistent.

The denominator is N , rather than $N - 1$, because the population mean is known to be zero.

Problem 02

Prove that the PDF of $\hat{A}$ given in Example 2.1 is $\mathcal{N}(A, \sigma^2/N)$.

Solution:

In Problem 01, the observations follow the model

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1,$$

where A is an unknown constant and

$$w[n] \sim \mathcal{N}(0, \sigma^2).$$

The estimator is the sample mean:

$$\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] = A + \frac{1}{N} \sum_{n=0}^{N-1} w[n].$$

A linear combination of independent Gaussian random variables is Gaussian. Consequently, $\hat{A}$ is Gaussian, and its distribution is completely determined by its mean and variance.

Its mean is

$$E[\hat{A}] = A + \frac{1}{N} \sum_{n=0}^{N-1} E[w[n]] = A.$$

Using the independence of the noise samples, its variance is

$$\begin{aligned} \text{Var}(\hat{A}) &= \text{Var}\left(\frac{1}{N} \sum_{n=0}^{N-1} w[n]\right) \\ &= \frac{1}{N^2} \sum_{n=0}^{N-1} \text{Var}(w[n]) \\ &= \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N}. \end{aligned}$$

Therefore,

$$\hat{A} \sim \mathcal{N}\left(A, \frac{\sigma^2}{N}\right)$$

Its probability density function is

$$f_{\hat{A}}(a; A) = \sqrt{\frac{N}{2\pi\sigma^2}} \exp\left[-\frac{N(a-A)^2}{2\sigma^2}\right], \quad -\infty < a < \infty.$$

This distribution is exact for every N ; it does not require a central limit theorem approximation. Strictly speaking, the problem concerns the PDF of $\hat{A}$, since A is a fixed unknown parameter.

Problem 03

For the problem described in Example 2.1 the more general estimator

$$\hat{A} = \sum_{n=0}^{N-1} a_n x[n]$$

is proposed. Find the a_n 's so that the estimator is unbiased and the variance is minimized. Hint: Use Lagrangian multipliers with unbiasedness as the constraint equation.

Solution:

Under the same observation model, consider the linear estimator

$$\hat{A} = \sum_{n=0}^{N-1} a_n x[n],$$

where the a_n 's are deterministic coefficients.

First, impose the unbiasedness requirement. Since $E[x[n]] = A$,

$$E[\hat{A}] = \sum_{n=0}^{N-1} a_n E[x[n]] = A \sum_{n=0}^{N-1} a_n.$$

For this expectation to equal A for every possible value of A , the coefficients must satisfy $\sum_{n=0}^{N-1} a_n = 1$. Because the observations are independent and each has variance σ^2 ,

$$\text{Var}(\hat{A}) = \sum_{n=0}^{N-1} a_n^2 \text{Var}(x[n]) = \sigma^2 \sum_{n=0}^{N-1} a_n^2.$$

The optimization problem is therefore

$$\min_{\{a_n\}} \sigma^2 \sum_{n=0}^{N-1} a_n^2 \text{ subject to } \sum_{n=0}^{N-1} a_n = 1.$$

Introduce the Lagrangian

$$\mathcal{L}(a_0, \dots, a_{N-1}, \lambda) = \sigma^2 \sum_{n=0}^{N-1} a_n^2 + \lambda \left(\sum_{n=0}^{N-1} a_n - 1 \right).$$

Differentiating with respect to each coefficient and setting the derivative to zero gives

$$\frac{\partial \mathcal{L}}{\partial a_n} = 2\sigma^2 a_n + \lambda = 0.$$

Hence,

$$a_n = -\frac{\lambda}{2\sigma^2},$$

so all coefficients must be equal. Substituting into the constraint yields

$$-\frac{N\lambda}{2\sigma^2} = 1 \Rightarrow \lambda = -\frac{2\sigma^2}{N}.$$

Therefore,

$$a_n = \frac{1}{N}, n = 0, 1, \dots, N-1.$$

For $\sigma^2 > 0$, the objective function is strictly convex and the constraint is linear, so this solution is the unique global minimum.

Thus, the minimum-variance linear unbiased estimator is the sample mean:

$$\hat{A}_{\text{opt}} = \frac{1}{N} \sum_{n=0}^{N-1} x[n],$$

with minimum variance

$$\text{Var}(\hat{A}_{\text{opt}}) = \sigma^2 \sum_{n=0}^{N-1} \left(\frac{1}{N}\right)^2 = \frac{\sigma^2}{N}.$$